Krusell–Smith Economies in Lean: Existence, Uniqueness, Stationarity and the Aggregate State

LeanEconomics Org [mostly Claude]*

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Abstract

This is the record of a programme to formalise heterogeneous-agent economies with aggregate risk in the style of Krusell and Smith [1998] in Lean 4, and of what it proves. Sequential equilibrium exists from every initial state, for logarithmic and CRRA utility and for Cobb–Douglas prices. Recursive equilibrium exists when the aggregate state is enlarged by the value function or by the infinite history of aggregate shocks, and a stationary recursive equilibrium with an ergodic distribution exists in the sense of Duffie et al. [1994]. When the capital share is small, sequential equilibrium is unique, so a recursive equilibrium exists on the minimal state (z, μ); the law of motion contracts, which turns the matched-expectations idea of Lee [2026] into explicit error bounds; and the equilibrium is stationary directly from the sequential equilibrium, with a unique and ergodic invariant distribution. We then ask when aggregate shock and aggregate capital (z, K) suffice as the state. We characterise the answer exactly: a recursive equilibrium on (z, K) exists if and only if two histories with the same (z, K) have the same next capital (under Krusell and Smith's employment design). We bound the error of forecasting with (z, K) by the nonlinearity of saving plus the feedback of prices, and give an explicit economy with heterogeneous households and Cobb–Douglas prices in which (z, K) is exactly sufficient: sequential equilibrium is unique, (z, K) is a sufficient state, and the recursive equilibrium on (z, K) has a unique, ergodic stationary distribution, with persistent aggregate shocks. A second economy, with a linear technology and i.i.d. shocks, has households saving strictly inside their constraints and a cross-section that keeps its whole history, and (z, K) is still sufficient. Whether (z, K) is sufficient at the interior, persistent calibrations of Krusell and Smith, with prices responding to capital, remains open.

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*The Lean development described here is at `https://github.com/LeanEconomics/LeanEconomics`, directory `LeanEconomics/KrusellSmith`. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result reported was checked by the Lean kernel, and every headline theorem depends only on Lean's standard axioms. Companion to the write-ups on Aiyagari (1994) and on overlapping generations in the same repository.

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1 Introduction

Krusell and Smith [1998] solve a heterogeneous-agent economy with aggregate risk by supposing that households forecast prices from the aggregate shock and the mean of the wealth distribution, and find that this forecast is almost exact. Behind the method lie questions the method does not answer. Does an equilibrium exist? Is it recursive in some state, and in which? Is the economy stationary? And when is the aggregate state (z, K) — shock and capital — actually enough?

This note records what a Lean formalisation of the model proves about these questions. The results are:

1. **Existence** (Section 3). From every aggregate shock and initial distribution there is a sequential equilibrium, by Schauder–Tychonoff on paths of capital over the tree of shock histories; with logarithmic and CRRA utility; with Cobb–Douglas prices, the capital clamp provably slack under an explicit condition.

2. **Recursive equilibrium** (Section 4). With the value function, or the infinite history of aggregate shocks, added to the state, a recursive equilibrium exists in every economy; a stationary recursive equilibrium with an ergodic distribution exists in the sense of Duffie et al. [1994], conditionally spotless in the sense of Cao [2020]. Uniqueness of sequential equilibrium implies a recursive equilibrium on the minimal state (z, μ).

3. **Uniqueness for small capital shares** (Section 5). For CRRA utility and Cobb–Douglas prices there is a capital share below which sequential equilibrium is unique, checked from primitives by a contraction in marginal utility. The law of motion then contracts in the distribution, which bounds the error of matching on capital [Lee, 2026].

4. **Stationarity from the sequential equilibrium** (Section 6). The equilibrium is a function of the infinite history of shocks, so it inherits stationarity from the shocks: the aggregate state of the unique small-share equilibrium has a unique, ergodic invariant distribution, with no recursive state space and no randomisation.

5. **The aggregate state** (z, K) (Section 7). A recursive equilibrium on (z, K) representing the equilibrium exists if and only if histories with the same (z, K) have the same future capital and labour; then it is continuous and its stationary law is unique. Without idiosyncratic heterogeneity this always holds. The error of forecasting next capital from (z, K) is at most the nonlinearity of saving plus the price feedback, times the distance between cross-sections.

6. **An explicit heterogeneous economy** (Section 8). Log utility, Cobb–Douglas prices, employed households saving to a cap and unemployed households spending everything: sequential equilibrium is unique from every initial state, (z, K) is sufficient, and the recursive equilibrium on (z, K) has a unique, ergodic stationary distribution. Aggregate shocks have Krusell and Smith's persistence.

7. **Interior saving** (Section 9). With prices depending on the shock only (a linear technology), i.i.d. shocks and cash ordered by the shock, (z, K) is sufficient in every sequence-state equilibrium; in an explicit economy half of the households save strictly inside their constraints after every good shock.

Section 10 lists what is open. Appendix A indexes the formal statements.

The prior literature is close in places. Miao [2006] proves existence of sequential equilibrium by Schauder–Tychonoff and a recursive equilibrium with payoffs in the state; Cao [2020] proves existence for the Krusell–Smith specification, a generalised recursive equilibrium, the value function as extended state and, under uniqueness, the minimal-state recursive equilibrium, and leaves the minimal-state question without uniqueness open; Kubler and Schmedders [2002] give finite-agent examples where the wealth distribution is not a sufficient state. Both Miao [2006] and Kubler and Schmedders [2002] treat uniqueness as a condition that cannot in general be verified from primitives; Section 5 verifies it from primitives, though only for small capital shares. Pröhl [2025] claims existence and uniqueness of a minimal-state recursive equilibrium by monotone-operator methods; we found gaps in the proofs as written

(the interiority argument in the existence step, and convexity of the admissible set) and do not rely on it.

2 The model as formalised

Aggregate shocks z take finitely many values and follow a Markov chain $\pi(z, z')$. Each household has an idiosyncratic state e in a finite set, moving with probabilities $p(z, z', e, e')$ that may depend on today's and tomorrow's aggregate shock, and labour efficiency $\epsilon(e)$. Households hold positions $a - b$ with $a \in [0, \bar{a}]$, where $b \geq 0$ is a borrowing limit, discount at $\beta < 1$, and receive the gross return $R(z, K, L)$ on their position and the wage $w(z, K, L)$ per efficiency unit, where K is mean (shifted) assets and L aggregate labour. Cash on hand is $R(a - b) + w\epsilon(e)$; saving a' is feasible when $0 \leq a' \leq \min(\bar{a},\ R(a - b) + w\epsilon(e) - \underline{c})$, with $\underline{c} \geq 0$ a consumption floor that labour income net of debt service at the limit always affords. Krusell and Smith's economy has $b = 0$; results that need it say so (the monotone route of Section 4 and the capital lower bound below, since a higher return hurts debtors). Period utility is continuous, increasing and strictly concave on $[\underline{c}, \infty)$.

Prices are abstract bounded continuous functions; the Cobb–Douglas instance is $R = 1 - \delta + \alpha A_z K^{\alpha-1} L^{1-\alpha}$, $w = (1 - \alpha) A_z K^{\alpha} L^{-\alpha}$ with capital and labour clamped to boxes, so that prices are bounded. Utility unbounded below at zero (logarithmic, CRRA with $\gamma \geq 1$) enters through a positive income floor and a consumption floor proved never to bind (`KSInada`). A household's problem along a path of prices is a stochastic dynamic program on the tree of shock histories; its value function is the fixed point of a Blackwell contraction, and strict concavity gives a unique optimal policy, jointly continuous in assets and the price path.

Definition 2.1. A *sequential equilibrium* from shock z_0 and distribution μ_0 is a path K of capital on the tree of shock histories such that, at every node, capital equals mean assets of the distribution generated by optimal saving along the price path of K (with labour the mean efficiency of the idiosyncratic marginal).

Two features of the formalisation should be kept in mind. The asset cap $\bar{a}$ is a constraint; for small capital shares it is proved irrelevant (Theorem 3.4). And the clamping boxes of the Cobb–Douglas instance are part of the price specification; where it matters we prove that the equilibrium stays inside them, so that prices are the true marginal products.

3 Existence of sequential equilibrium

Theorem 3.1. *In every Krusell–Smith economy, from every aggregate shock z_0 and every initial distribution μ_0, a sequential equilibrium exists, with aggregate labour equal to the mean efficiency at every node.*

Formal statement: `exists_sequentialEquilibrium, exists_sequentialEquilibrium_labour`.

The unknown is a path of capital on the tree of shock histories, with values in $[0, \bar{a}]$. These paths form a compact convex subset of a locally convex space in the product topology. The

household value function is continuous in the path: a depth-weighted sup norm makes the Bellman operator a contraction with a modulus uniform in the path, and the weights make the tree's distant nodes negligible. Strict concavity gives a unique policy, continuous in the path by Berge's theorem; the distributions on the tree are then weakly continuous in the path by induction on depth, and so is mean capital. Schauder–Tychonoff gives a fixed point.

Theorem 3.2. *The same holds with utility defined only on* $(0, \infty)$ *and marginal utility tending to infinity at zero — logarithmic and CRRA utility — provided labour income is bounded below by a positive floor; optimal consumption never reaches the consumption floor used in the construction.*

Formal statement: `KSInada.exists_sequentialEquilibrium`.

Theorem 3.3. *With logarithmic utility and Cobb–Douglas prices, a sequential equilibrium exists, and prices are the true marginal products wherever capital lies in the clamping box. With CRRA utility, an explicit lower end* $\underline{K}$ *of the capital box (depending on* β*,* γ*, the persistence of the aggregate chain and the technology) keeps every equilibrium's capital above* $\underline{K}$*: the clamp never binds.*

Formal statement: `logEconomy_exists_equilibrium, crra_exists_unclamped_equilibrium`.

The lower bound adapts Cao [2020, Lemma 9]: a per-household Euler inequality, aggregated with Jensen's inequality for the convex function $c \mapsto c^{-\gamma}$, bounds consumption today by expected consumption at the child nodes, and an induction on the tree pushes capital above $\underline{K}$.

The asset cap. The compact asset set $[0, \bar{a}]$ is what Schauder–Tychonoff needs, but the cap can be shown irrelevant. When saving rises with assets at most at slope θ — with CRRA utility, the minimal-MPC slope $\theta = \bar{R}(\beta\bar{R})^{1/\gamma}/\underline{R}$ of Section 5, which holds whether or not the cap binds — the poorest household saves at most its labour income, so every household saves at most $\bar{w}e + \theta a$. If $\theta < 1$, saving never reaches a cap above $\bar{w}e/(1-\theta)$, a threshold that does not involve the cap. Moreover, shrinking the cap to any $\bar{a}_0$ with $\bar{w}e + \theta\bar{a}_0 \le \bar{a}_0$ changes nothing: the smaller economy's value function and policy are the restrictions of the larger one's (the maximum over the larger feasible set is attained in the smaller), distributions on $[0, \bar{a}_0]$ move the same way, and the sequential equilibria from them coincide.

Theorem 3.4. *For CRRA utility and Cobb–Douglas prices with* $0 \le \delta < 1$ *and* $(\beta(1-\delta))^{1/\gamma} < 1 - \delta$*, below some capital share:* $\theta < 1$*; saving is strictly below any cap exceeding* $\bar{w}e/(1-\theta)$*, at every state and along every price path; and every cap* $\bar{a}_0 \ge \bar{w}e/(1-\theta)$ *gives the same sequential equilibria from distributions on* $[0, \bar{a}_0]$.

Formal statement: `minMPC_smallShare_capIrrelevant, policy_lt_assetCap_of_slope, isSequentialEquilibrium_capEmb_iff, policy_capEmb, valueFunction_capEmb`.

Outside the small-share class the cap remains an assumption: $\theta < 1$ needs the spread of returns $\bar{R}/\underline{R}$ to be below $(\beta\bar{R})^{-1/\gamma}$. The consumption floor of the CRRA model depends on the cap through a value bound, so the comparison is between the economy and its own smaller-cap versions; the floor is slack in both.

Borrowing. With a borrowing limit $b > 0$ every existence result above goes through, with the income floor replaced by labour income net of debt service at the limit. For CRRA utility and Cobb–Douglas prices of actual capital (mean position $K - b$), a sequential equilibrium exists from every initial state whenever $b < \underline{w}\,\underline{\epsilon}/\bar{R}$, the analogue of a natural borrowing limit: labour income always covers the interest on the maximal debt.

Theorem 3.5. *For CRRA utility, clamped Cobb–Douglas prices of actual capital, and $0 \le b < \underline{w}\,\underline{\epsilon}/\bar{R}$, from every aggregate shock and distribution of positions there is a sequential equilibrium in which households borrow up to b, with consumption bounded away from zero.*

Formal statement: `crraBorrow_exists_sequentialEquilibrium`, `crraBorrowEconomy`, `KSModel.borrowLimit`.

4 Recursive equilibrium

4.1 Uniqueness gives the minimal state

Theorem 4.1. *If sequential equilibrium is unique from every (z, μ), then there is a recursive equilibrium on the minimal state (z, μ): the distribution next period is a function of today's shock, today's distribution and tomorrow's shock, weakly continuous in the distribution, and saving is a continuous function of today's state.*

Formal statement: `KSModel.isRecursive`, `continuous_lawOfMotion`.

The continuation of a sequential equilibrium below a node is a sequential equilibrium from that node's state, and the graph of the equilibrium correspondence is closed; with uniqueness the equilibrium path is a continuous function of (z, μ) by the closed-graph theorem on a compact space, and the law of motion is read off it. This is Cao [2020, Proposition 2].

4.2 Enlarged states, in every economy

Theorem 4.2. *In every economy there is a recursive equilibrium whose state is (z, μ, v), with v the profile of household values, and every such equilibrium generates a sequential equilibrium with the same payoffs.*

Formal statement: `recursiveEquilibrium_valueState`, `generates_sequentialEquilibrium`, `payoffEquivalent`.

Theorem 4.3. *In every economy there is a recursive equilibrium whose aggregate state is the infinite history of aggregate shocks: an assignment μ^* of a distribution to each history, consistent with optimal saving at every history, with the history evolving exogenously. This holds in particular with logarithmic utility and Cobb–Douglas prices.*

Formal statement: `exists_seqStateEquilibrium`, `exists_historyRecursiveEquilibrium`, `logEconomy_exists_seqStateEquilibrium`.

The fixed point is taken in the space of assignments of distributions to infinite histories; no initial distribution is needed. We call it the *sequence-state equilibrium*. Under uniqueness of sequential equilibrium it is Markov in (z, μ).

Theorem 4.4. *In every economy there is a stationary recursive equilibrium with an ergodic invariant distribution, on the expectations-augmented state of Duffie et al. [1994]; it can be chosen conditionally spotless in the sense of Cao [2020, Definition 6].*

Formal statement: `exists_stationaryRecursiveEquilibrium, exists_conditionallySpotlessEquilibrium`.

The proof formalises DGMM's Theorem 1.1 and Corollary 1.1 in the abstract (Kakutani–Fan–Glicksberg for a correspondence of measures, disintegration, Krein–Milman for the ergodic choice) and applies it to the compact set of equilibrium states.

4.3 The minimal state without uniqueness

Theorem 4.5. *If saving rises with the path of aggregate capital, the greatest sequential equilibrium is a consistent selection, and a recursive equilibrium on (z, μ) exists without uniqueness. Saving rises with capital in complementarity economies (a constant wage, a return nondecreasing in capital, relative risk aversion at most one), and an explicit such economy is given.*

Formal statement: `SavingMonotone.isRecursive, savingMonotone_of, example_exists_consistentSelection`.

The condition fails for Cobb–Douglas prices, where higher capital lowers the cash on hand of wealthy households.

5 Uniqueness for small capital shares, and matching

5.1 A contraction in marginal utility

Uniqueness is proved by showing that the equilibrium map — capital path to the mean-saving path it induces — is a contraction in a depth-weighted sup norm. The household side is measured in marginal utility, not in policies: the gap in $u'(c)$ between two price paths propagates with factor $\beta\bar{R}$ per period, and the policy gap follows by strong concavity on the range of consumption. With ℓ the Lipschitz constant of prices in capital, the contraction factor is $\kappa = \lambda(\ell C + A/m)/(1 - \lambda\theta)$, explicit in the primitives, where θ is a bound on the slope of saving in assets.

Theorem 5.1. *For CRRA utility ($\gamma > 0$) and Cobb–Douglas prices with the clamp, there is $\alpha_0 > 0$ such that for every capital share $\alpha < \alpha_0$ sequential equilibrium is unique from every state, so the recursive equilibrium on (z, μ) of Theorem 4.1 exists. This holds for $0 \le \delta < 1$ whenever $(\beta(1-\delta))^{1/\gamma} < 1 - \delta$, using the slope bound $\theta = \bar{R}(\beta\bar{R})^{1/\gamma}/\underline{R}$ obtained from the minimal marginal propensity to consume; in particular for $\delta = 0$ unconditionally.*

Formal statement: `crra_existsUnique_sequentialEquilibrium, smallShare_contraction, minMPC_smallShare`.

The proof is analytic: κ and the rate of the law of motion are explicit functions of α, continuous at $\alpha = 0$ where they reduce to 0 and to the slope bound. The theorem is qualitative. At the Krusell–Smith calibration the crude constants give κ of order 10^{15} at $\alpha = 0.36$, and $\kappa < 1$

only for capital shares of order 10^{-13} (a separate numerical evaluation, not part of the formal development).

Theorem 5.2. *Under the same conditions, if employment transitions are bounded below, there is exactly one sequence-state equilibrium.*

Formal statement: `minMPC_smallShare_existsUnique_seqStateEquilibrium, minor_minMPC_smallShare_existsUnique_seqStateEquilibrium`.

5.2 Fading memory and matched expectations

Lee [2026] argues that matching on aggregate capital along a simulated path is accurate because the economy forgets its past. In the small-share class this is a theorem. Measure distance between cross-sections by a two-part distance: Lipschitz test functions in assets, and the employment marginals.

Theorem 5.3. *For small capital shares, with both shock chains bounded below, the equilibrium law of motion contracts at a rate $r^* < 1$ in the two-part distance, even when employment marginals depend on the history; in every sequence-state equilibrium, two histories agreeing over their last n shocks have savings within $C_0(r^*)^n$, with C_0 explicit.*

Formal statement: `dist2_run, minMPC_smallShare_matching, smallShare_matching`.

6 Stationarity directly from the sequential equilibrium

The usual route to a stationary equilibrium passes through a recursive state space, often enlarged and randomised [Duffie et al., 1994]. A sequence-state equilibrium offers a direct one: it is a function of the infinite history of shocks, and the history evolves exogenously, so if the history is drawn from a law left invariant by the shock process, the aggregate state (z, μ^*) has the same law every period. The equilibrium is a factor of the stationary shock process.

Theorem 6.1. *The shock-history process has an invariant ergodic law, unique when the aggregate chain is bounded below. For every sequence-state equilibrium measurable in the history — with no uniqueness — tomorrow's aggregate state has today's law. Under uniqueness of sequential equilibrium that law is an invariant distribution of the equilibrium Markov process on (z, μ), ergodic when the shock law is.*

Formal statement: `exists_invariant_ergodic, stationary_unique_of_minor, stateLaw_tomorrow, isInvariant_stateLaw`.

Theorem 6.2. *For CRRA utility, Cobb–Douglas prices with $0 \le \delta < 1$ and $(\beta(1-\delta))^{1/\gamma} < 1-\delta$, both shock chains bounded below and a small enough capital share: the sequence-state equilibrium is unique and continuous in the history; the law of its aggregate state under the stationary shock law is an invariant and ergodic distribution of the equilibrium Markov process; and every invariant distribution of that process equals it.*

Formal statement: `continuous_seqState, minMPC_smallShare_unique_stationary`.

Continuity comes from fading memory (Theorem 5.3) and Stone–Weierstrass density of asset-Lipschitz test functions. Uniqueness of the invariant distribution is a coupling: the T-step expectation of a test function from any state is within $C(r^*)^T$ of a function of today's shock alone, invariant distributions have the same shock marginal, and the test functions separate measures on the Polish aggregate state space. The theorem proves uniqueness of the invariant distribution, not convergence in law from every starting point.

7 When is (z, K) a sufficient state?

Households need only prices, and prices depend on the path of capital and labour. So the right question is not whether (z, K) determines the distribution, but whether it determines that path.

Definition 7.1. A sequence-state equilibrium μ^* satisfies (z, K)-*sufficiency* if any two histories with the same current shock and the same capital have the same capital and labour at every node of their future trees.

A *recursive equilibrium on* (z, K) consists of capital sets $\mathcal{K}_z$, a law of motion $K' = \Psi(z, K, z')$ keeping capital in the sets, labour $L(z, K)$, and value and saving functions of (z, K, a, e) satisfying the Bellman equation with continuation at $(z', \Psi(z, K, z'))$, together with consistency of aggregation: each (z, K) carries a nonempty set of distributions with mean K and labour $L(z, K)$, mapped by the saving policy and the employment transitions into the set at $(z', \Psi(z, K, z'))$. It *represents* μ^* if at every history the capital lies in $\mathcal{K}_z$, the distribution lies in the set at (z, K), and households save by the (z, K) policy.

Theorem 7.2. *For every sequence-state equilibrium, (z, K)-sufficiency holds if and only if some recursive equilibrium on (z, K) represents it. For small capital shares (as in Theorem 6.2): when (z, K)-sufficiency holds, the capital sets are compact, the law of motion Ψ is continuous on them, and the law of (z, K) is the unique invariant distribution of the (z, K) process on the capital sets, and it is ergodic.*

Formal statement: `kSufficient_iff_exists_represents, minMPC_smallShare_aggregateState, stationary_aggLaw`.

Continuity needs no injectivity: capital is a quotient map from the compact set of histories with a given shock. Uniqueness of the stationary law needs no Lipschitz test functions and no measurable selection of distributions: bounded continuous test functions are uniformly continuous on the compact capital sets, and capital forgets old shocks.

Theorem 7.3. *With a single idiosyncratic state, every household holds aggregate capital in the small-share equilibrium, so capital determines the distribution and (z, K)-sufficiency holds: the stochastic growth model of Brock and Mirman [1972] is recovered as a recursive equilibrium on (z, K) with a unique stationary distribution.*

Formal statement: `minMPC_smallShare_repAgent`.

An order route. If the cross-sections the equilibrium visits at each shock form a chain for first-order stochastic dominance in assets, then capital determines the cross-section, because two comparable cross-sections with the same mean are equal. The chain property follows from two conditions: the law of motion preserves dominance, and after ordered shocks the next cross-sections are ordered whatever the pasts. The first holds, by Tarski comparative statics, when saving rises with the capital path (Theorem 4.5); the second reduces to finitely many comparisons between extreme histories.

Theorem 7.4. *Dominance with equal means is equality. If saving rises with the capital path, sequential equilibrium is unique, the equilibrium is continuous in the history, a better newest shock raises the cross-section, and the extreme comparisons hold, then (z, K)-sufficiency holds.*

Formal statement: `eq_of_dominates_of_mean_eq, kInjective_of_chain, kSufficient_of_savingMonotone, kSufficient_of_extremes`.

The extreme comparisons say that one period of a better shock outweighs a permanent boom against a permanent bust. They force capital to encode the whole shock history, which needs the economy to forget at least half of a capital deviation each period with two shocks. They cannot hold at the persistence of Krusell and Smith's calibration, which says nothing about (z, K)-sufficiency there, only that this sufficient condition is unavailable.

Approximate aggregation. When (z, K) is not exactly sufficient, how wrong is it? Sufficiency is equivalent to the one-step forecast being exact: if histories with the same (z, K) always have the same next capital, the whole future price tree agrees by induction (labour is pinned by the shock). So the one-step forecast error is the quantity to bound. Two cross-sections with the same mean assets integrate a function whose slopes in assets lie in $[\theta_{lo}, \theta_{hi}]$ to within $\frac{1}{2}(\theta_{hi} - \theta_{lo})\, d$, where d is their Wasserstein distance in assets (within employment states): subtract the mid-slope times assets, which the equal means cancel. Applied to saving, and adding the effect of the two cross-sections facing different future prices, next capital differs by at most

$$\varepsilon\, d, \qquad \varepsilon = \frac{\theta - \theta_{lo}}{2} + \frac{G}{1 - \kappa},$$

with θ the saving slope of Theorem 5.1, θ_{lo} any lower slope ($\theta_{lo} = 0$ always, since saving is monotone), and $G/(1 - \kappa)$ the price feedback. The common slope of saving drops out: what matters is its nonlinearity across the cross-section, not how much households save.

Theorem 7.5. *(z, K) is sufficient if and only if the one-step forecast of capital from (z, K) is exact. In the small-share class with Krusell and Smith's employment design, every sequence-state equilibrium has a law of motion $\ell(z, K)$ with $|K_{t+1} - \ell(z_t, K_t)| \le \varepsilon \cdot 2\bar{a}$ at every history; two histories with the same (z, K) that agree over their last $T + 1$ shocks have next capital within $\varepsilon\, r^T\, 2\bar{a}$.*

Formal statement: `kSufficient_of_forecast_exact, abs_integral_sub_le_of_band, SlopeContraction.capital_gap, SlopeContraction.approxAggregation, minMPC_smallShare_approxAggregation, SlopeContraction.forecast_gap_of_agree`.

The bound vanishes for affine saving with prices that do not respond to capital, and it names the mechanism usually offered for Krusell and Smith's finding: saving is nearly linear in wealth over most of the cross-section, and capital moves prices little. It does not make the error small at their calibration: saving is flat at the borrowing constraint, so $\theta - \theta_{lo}$ is not small, and the bound uses the whole asset range rather than the cross-sections the equilibrium visits.

8 An explicit economy with heterogeneous households

8.1 Corner economies

Let the employed save up to the cap and the unemployed spend everything. Under the employment design of Krusell and Smith [1998] — the employment rate at shock z is $m(z)$, consistent under the transitions — the cross-section after yesterday's shock y and today's z puts mass on two asset levels, $\bar{a}$ for those employed last period and 0 for the others, jointly with the current employment state. Capital is $\bar{a}\, m(y)$. Below the root of a history's tree, capital and labour depend only on the shocks along the node and today's shock, so yesterday's shock enters prices only through today's capital: (z, K)-sufficiency holds with no injectivity.

The corner policy is optimal when three conditions hold along the price path: labour income covers the floor plus the corner saving; for the employed, $u'(c) \leq \beta\mathbb{E}[R'u'(c')]$ at the cap; for the unemployed, $u'(c) \geq \beta\mathbb{E}[R'u'(c')]$ at zero, where c' is consumption next period under the corner policy itself. The proof is a verification argument: the value of following the corner policy is today's utility plus a term independent of assets; against it the objective is concave, the tangent lines today and tomorrow bound the gain from leaving the corner by $(g - a)\,(u'(c) - \beta\mathbb{E}R'u'(c'))$, which has the right sign; so the policy's value solves the Bellman equation and is the value function. With prices bounded in $[R_{lo}, R_{hi}] \times [w_{lo}, w_{hi}]$ and marginal utility decreasing, the conditions hold along every price path once they hold at their worst cases.

Formal statement: `policy_eq_cornerPolicy, policyValue_eq_valueFunction, cornerConditions_of_bounds`.

8.2 The economy

Aggregate shocks	two, persistent: each lasts with probability 7/8
Employment	efficiency 2 (employed), 1/4 (unemployed)
Employment rates	4/5 (bad), 17/20 (good); job finding 49/50
Job loss	49/200, 73/400, 197/850, 147/850 by (yesterday, today)
Preferences	logarithmic, $\beta = 24/25$; asset cap $\bar{a} = 1/16$
Technology	Cobb–Douglas, $\alpha = 1/10$, TFP 1/4, $\delta = 1/10$
Boxes	$K^{1/10} \in [11/15,\ 3/4]$, $L^{1/10} \in [21/20,\ 16/15]$

The box endpoints are tenth powers of rationals, so the powers 1/10 and 9/10 in the prices are exact and every inequality is checked in exact rational arithmetic. Gross returns lie in

about $[1.42, 1.63]$ and wages in about $[0.155, 0.161]$. Capital is $1/20$ after a bad shock and $17/320$ after a good one, labour $33/20$ or $139/80$: both inside the boxes at every node of the equilibrium price tree, so prices are the true Cobb–Douglas marginal products.

Theorem 8.1. *In this economy: sequential equilibrium is unique from every initial state; the sequence-state equilibrium exists, is unique, and is the corner one; (z, K)-sufficiency holds and a recursive equilibrium on (z, K) represents it; and its law of motion on (z, K) has a unique stationary distribution on the capital sets — weight $\frac{1}{2}\pi(y, z)$ on today's shock z with capital from yesterday's shock y, so $7/16$ where the shock persisted and $1/16$ where it switched — which is ergodic.*

Formal statement: `cornerUnique, cornerWitness, cornerStationary, ergodic_statLaw, seqPrices_mem_box`.

Uniqueness comes cheaply because the corner conditions hold along every price path: the household's policy is the corner one whatever it expects, so the distribution path does not depend on prices. The aggregate shocks have the persistence of Krusell and Smith's calibration (expected spells of eight periods); the corner conditions are checked at worst-case prices, so persistence costs nothing. Three features limit the example. Assets sit at corners rather than at interior precautionary levels (Section 9 removes this one, at other costs). The gross return exceeds $1/\beta$, which is what makes the cap bind for the employed. And with capital share $1/2$ and labour inside its box the corner economy cannot exist: the employed's net income over the unemployed's spending is bounded by roughly $(1 - \alpha)/\alpha$, too small for both corner conditions.

9 An economy with interior saving

The corner economy keeps the cross-section on two asset levels, so its history is short. Here the cross-section carries its whole history, saving is interior, and (z, K) is still sufficient, because capital encodes the history. The route is the order route of Section 7, with an economy in which its hypotheses can be checked.

9.1 Separation from ordered cash

Three conditions make the two hard hypotheses of Theorem 7.4 free.

- **Prices depend on the shock only**, as with a linear technology $Y = A(z)K + B(z)L$, whose marginal products are exact. Then the household's problem does not depend on the path of capital, so saving rises (weakly) with capital, and sequential equilibrium is unique: the distribution path does not depend on prices.
- **Shocks are i.i.d.**: next period's aggregate shock and employment are drawn independently of today's. Then the value of next period's assets is the same function at the root of every history, so today's saving is one nondecreasing function of cash on hand (a Topkis argument across histories).

- **Cash is ordered by the shock**: after a worse shock every household has less cash than every household after a better one.

Under these, every household saves less after the worse shock, so the next cross-sections are ordered for any two current ones. This is the separation condition of the order route, directly, without the extreme histories.

Theorem 9.1. *If prices depend on the shock only, shocks are i.i.d. and cash on hand is ordered by the shock, then in every sequence-state equilibrium (z, K) is a sufficient state.*

Formal statement: `kSufficient_of_orderedCash, sepOnHistories_of_orderedCash, policy_le_policy_of_cont, cont_root_eq, uniqueSequentialEquilibrium_of_exogenous`.

9.2 Interior saving

Interiority is proved by two corner deviations. Extra assets δ are worth at least $R\delta/(\text{cash}+R\delta)$ next period: keep the saving of the poorer state and consume the extra return, and use the tangent line of the logarithm. The last δ below b is worth at most $R\delta/(\text{cash}(b) - \bar{a} - R\delta)$: the poorer state copies the richer one's saving. So saving is positive at cash x if

$$\frac{\delta}{x-\delta} < \beta\,\mathbb{E}\left[\frac{R'\delta}{w'e' + R'\delta}\right],$$

and below the cap if

$$\frac{\delta}{x-\bar{a}+\delta} > \beta\,\mathbb{E}\left[\frac{R'\delta}{(R'-1)\bar{a} + w'e' - R'\delta}\right].$$

Formal statement: `value_sub_ge, value_sub_le, policy_pos_of, policy_lt_cap_of`.

9.3 The economy

Aggregate shocks	two, i.i.d. with probability 1/2
Efficiency	1 or 3/4, i.i.d. with probability 1/2
Technology	linear: gross return 6/5; wage 1 (good), 1/3 (bad)
Preferences	logarithmic, $\beta = 1/2$; asset cap $\bar{a} = 25/72$; floor 1/100

After a bad shock cash is at most $\frac{6}{5}\cdot\frac{25}{72} + \frac{1}{3} = \frac{3}{4}$; after a good one it is at least $\frac{3}{4}$. With $\delta = 1/3000$, for low-efficiency households after a good shock (cash in [3/4, 7/6]) the first condition reads $1/2249 < \frac{1}{8}(1/2501 + 1/1876 + 3/2503 + 1/626)$ and the second $3/7378 > \frac{1}{8}\cdot 18\,(1/48107 + 1/36857 + 1/18107 + 1/14357)$, both in exact rational arithmetic.

Theorem 9.2. *This economy has a sequence-state equilibrium. In every one, (z, K) is a sufficient state, and after every good shock at least half of the households — all those with low efficiency, whatever their assets — save strictly between 0 and $\bar{a}$.*

Formal statement: `InteriorWitness.exists_interiorWitness, InteriorWitness.interiorWitness, InteriorWitness.saving_interior, InteriorWitness.orderedCash`.

The costs are plain. Prices do not respond to capital, so the equilibrium feedback that makes Krusell–Smith hard is absent; what remains is the aggregation question itself, whether aggregate capital pins down a cross-section that keeps its whole history, and here it does. Shocks are i.i.d. rather than persistent. And the household is impatient: with these crude interiority bounds the windows are tiny for $\beta \geq 0.8$, so the period should be read as long. We do not prove that interior saving is strictly increasing in assets, which would make the cross-section infinitely supported.

10 What is open

1. (z, K) **at interior, persistent calibrations with price feedback.** The corner economy of Section 8 has persistent shocks and Cobb–Douglas prices but assets at corners; the economy of Section 9 has interior saving but i.i.d. shocks, prices that do not respond to capital, and an impatient household. Individual nonlinearity of saving does not by itself defeat (z, K)-sufficiency — only the cross-sections the equilibrium visits matter — so the question is open in both directions at Krusell and Smith's calibration. Price feedback inside the order route needs saving to rise with the capital path (which fails for Cobb–Douglas) and a uniqueness theorem for complementarity economies.

2. **Small forecast errors at realistic parameters.** Theorem 7.5 bounds the error of forecasting with (z, K) by the nonlinearity of saving and the price feedback, over the whole asset range. Making it small needs the nonlinearity measured only on the cross-sections the equilibrium visits, where few households are near the constraint.

3. **Measurable equilibria beyond small shares.** Stationarity directly from the sequential equilibrium needs the sequence-state equilibrium to be measurable in the history; outside the small-share class the fixed point gives no such guarantee. The natural fixed-point space — measurable maps into distributions, with the weak-* topology — is compact and convex, and capital at each node is continuous on it; but the household's policy is nonlinear in the whole price path, which converges only weakly in the history, so the update is not continuous. Running the economy from the far past and taking limits fails for the same reason. A contraction (as for small shares) or randomisation (as in Duffie et al. [1994]) removes the obstruction; a pure measurable equilibrium without either needs a quantitative modulus of the equilibrium price path in the history.

4. **The minimal state without uniqueness, for Cobb–Douglas.** The greatest-equilibrium route needs saving to rise with the capital path, and with Cobb–Douglas prices it cannot in any order. A rise in capital moves a household's cash at the rate $\alpha(1-\alpha)AK^{\alpha-2}L^{-\alpha}(Ke - La)$, up for the asset-poor and down for the asset-rich, and the change integrates to zero over the cross-section (Euler's theorem): capital only redistributes cash.[1] So a one-node perturbation of capital raises some households' saving and lowers others'; and when saving is affine in cash the redistribution washes out, leaving

[1] `CobbDouglasRedistribution.cash_deriv`, `CobbDouglasRedistribution.integral_redistribution_eq_zero`.

future-price effects that both lower saving — the equilibrium map reverses order. An order-reversing map brackets the equilibria between the least and greatest fixed points of its square; if the square is concave at the top of the bracket in Krasnosel'skii's sense and the bottom is a positive fraction of the top, the bracket closes, the equilibrium is unique and a minimal-state recursive equilibrium exists. That implication is formalised;[2] whether the Krusell–Smith map reverses order and has a concave square is open. A first obstacle: the zero-sum redistribution holds only where capital equals mean assets, that is at equilibrium, not along the off-equilibrium paths the map acts on. Pricing today at actual capital removes it (an expectations map with the same fixed points), and at the household level the needed sign holds: for CRRA with $\gamma \leq 1$, a higher expected capital path — lower future returns, higher future wages — lowers every household's saving at given cash.[3] What remains is propagation: below the first period the two cross-sections differ, today's prices redistribute cash between them, and an order on the cross-sections that survives this is again what is missing. This is the question Cao [2020] leaves open.

5. **Uniqueness at realistic capital shares.** Theorem 5.1 is qualitative; sharper bounds on the consumption floor and on curvature would be needed for a quantitative threshold.

6. **Full spotlessness**, open also in Cao [2020]; and the asset cap outside the small-share class (Theorem 3.4 needs the minimal-MPC slope below one).

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[2] `isRecursive_of_antitone`, `eq_of_concave`.
[3] `expMap_eq_iff`, `crra_policy_le_of_capital`.

Elisabeth Pröhl. Existence and uniqueness of recursive equilibria with aggregate and idiosyncratic risk. Working paper, 2025.

A Index of formal statements

All names are in the namespace `LeanEconomics.KrusellSmith` (most in `KSModel`); files are in `LeanEconomics/KrusellSmith/` unless noted.

Result	Formal name	File
Existence (Thm 3.1)	`exists_sequentialEquilibrium`	`Equilibrium`
Labour consistency	`exists_sequentialEquilibrium_labour`	`Labour`
Log, CRRA (Thm 3.2)	`KSInada.exists_sequentialEquilibrium`	`Inada`
Cobb–Douglas (Thm 3.3)	`logEconomy_exists_equilibrium`	`CobbDouglas`
Clamp never binds	`crra_exists_unclamped_equilibrium`	`CobbDouglasCapital`
Borrowing (Thm 3.5)	`crraBorrow_exists_sequentialEquilibrium`	`Borrowing`
Asset cap irrelevant (Thm 3.4)	`minMPC_smallShare_capIrrelevant`, `isSequentialEquilibrium_capEmb_iff`	`AssetCap`
Minimal state under uniqueness (Thm 4.1)	`KSModel.isRecursive`	`Recursive`
Value-function state (Thm 4.2)	`recursiveEquilibrium_valueState`	`ValueStateExistence`
History state (Thm 4.3)	`exists_seqStateEquilibrium`	`SequenceState Equilibrium`
DGMM stationary (Thm 4.4)	`exists_stationaryRecursiveEquilibrium`	`StationaryEquilibrium`
Conditionally spotless	`exists_conditionallySpotlessEquilibrium`	`SpotlessEquilibrium`
Monotone route (Thm 4.5)	`SavingMonotone.isRecursive`	`GreatestEquilibrium`
Uniqueness, CRRA (Thm 5.1)	`crra_existsUnique_sequentialEquilibrium`	`CRRAContraction`
Small capital shares	`minMPC_smallShare`	`MinMPCSmallShare`
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Result	Formal name	File
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